

Review

Bioengineering Parameters for Single-Use Bioreactors: Overview and Evaluation of Suitable Methods

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During the past five years, the number of single-use bioreactors used in biopharmaceutical research and production has increased tremendously. This increase has been particularly associated with mammalian cell culture processes from small- to medium-scale volumes. Even though nowadays customers can choose from a multitude of 2nd and 3rd generation single-use bioreactors, ranging from mL- up to m³-scale, there is a lack of knowledge of their engineering parameters. Different approaches have been applied to characterization investigations, resulting in an inability to compare different single-use bioreactors with each other and their reusable counterparts, creating an obstacle to a systematic approach to scaling-up the process. This article describes parametric, experimental and computer-based numeric methods for biochemical engineering characterization of single-use bioreactors, which have already been used successfully for the characterization of their reusable counterparts. For the first time, these methods have been evaluated in terms of their practical application.

Keywords: Computational fluid dynamics, Engineering characterization, Mixing time, Particle stress, Power input, Single-use bioreactor, Volumetric oxygen transfer coefficient

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1 Introduction

The bioreactor is a key element in any biopharmaceutical production process and is currently most often animal cell-based. Together with the cell line, cell culture medium, and process mode, the bioreactor represents the principal starting point for process optimization. Irrespective of the bioreactor type, its design and the resulting biochemical engineering specifications influence the cell growth of the production cell line [1] as well as the product quantity and quality [2, 3]. Although a multitude of bioreactor types have been developed during the last 60 years, only a few types of bioreactors prevail in today's production processes, among them are bubble columns, airlift reactors and stirred bioreactors made of stainless steel (which are also referred to as conventional bioreactors) [4].

Special design features, including a cultivation vessel height to diameter ratio of 2:1 or 3:1 and implementation of agitators with distributed power input, preferably magnetic agitators, centric agitators with baffles or angular agitators without baffles, have become accepted during the past 50 years [5–9]. These features are based on empirical data from industry as well as comprehensive biochemical engineering and cell biology studies. The engineering characterization for the bioreactors was performed under parametrical, experimental and numeric methods, which all originate from traditional biochemical engineering investigations (see Fig. 1). These methods allow mixing properties, oxygen transfer, fluid flow pattern, temperature and pressure distribution within bioreactors (independent on the type) to be determined. In addition, the determination of a bioreactor's specific power input, tip speed and energy dissipation rate,

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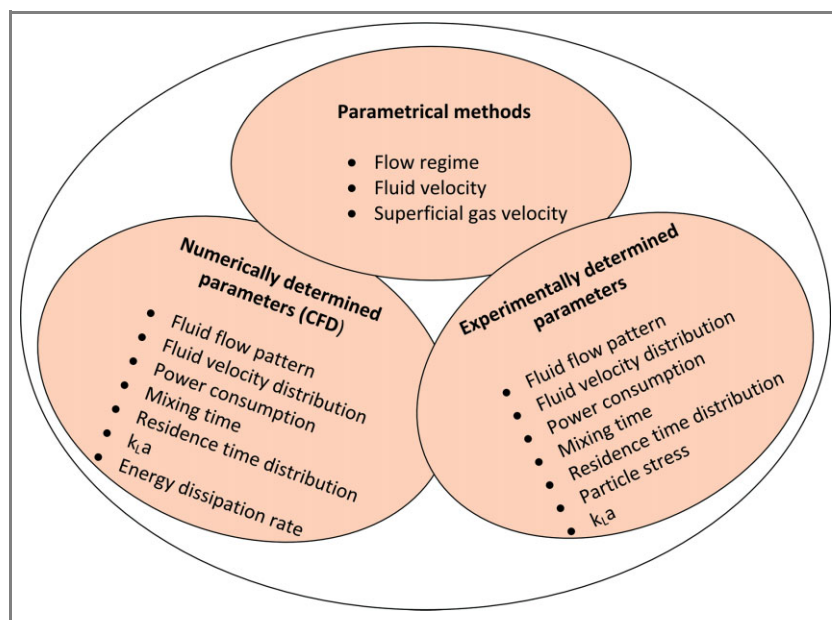


Figure 1. Bioengineering parameters for single-use bioreactors, which are determined by parametrical, experimental and numerical methods.

and thus estimation of shear stress, becomes possible. Finally, knowledge of a bioreactor's engineering parameters, such as stirrer speed, tip speed, mixing time, k_La value and specific power input, allow bioreactors to be quickly scaled-up by keeping one or two parameters constant [6, 10].

The first scalable single-use bioreactor developed for animal cells at the bench top scale, the "Wave", was launched in the late 1990s. But biochemical engineering data for this bioreactor and its successors were not available until mid-2000s. The reasons for this are twofold: this bioreactor has a flexible plastic cultivation bag instead of a plastic rigid vessel; and secondly, the wave induced motion (induction of a wave caused by the rocking of the bioreactor platform shaping and fixing the cultivation bag) presented an entirely new kind of mixing principle [11]. However, due to convincing cultivation results in the Wave, wave-mixed bioreactors have driven the development of further single-use bioreactors [12].

Based on a short overview of commercially available single-use bioreactors, suitable methods for their engineering characterization are discussed in the following sections. These methods have been developed for conventional systems and will be modified and transferred to single-use bioreactors. Finally, a recommendation for the advantageous investigation of the dominant stirred, wave-mixed and orbitally shaken single-use bioreactor systems is provided.

2 Single-Use Bioreactor Overview

In contrast to reusable bioreactors made from glass or stainless steel, single-use bioreactors are made of FDA-approved plastics [12, 13]. Although, their use results in numerous advantages, single-use bioreactors do have their limitations [13].

Issues most often discussed are the lack of instrumentation for single-use sensors [12] and possible secretion of leachables and extractables from the internal layer of the plastic bag or vessel, which is in direct contact with the product [14–18]. Nevertheless, users can choose from a multitude of single-use bioreactors today. As shown in Fig. 2 and Tab. 1, static single-use bioreactors on mL-scale (screening-scale) and dynamic systems up to m³-scale are available [19]. Static systems such as T-flasks are used for work routinely done in the lab. Dynamic single-use bioreactors are mechanically driven, pneumatically driven or so-called hybrid systems (combinations of both driven methods) ensuring higher cell densities and product titers in comparison to static systems.

As obvious from Tab. 1, which provides an overview of single-use bioreactors that are commercially available till this day, stirred, wave-mixed and orbitally shaken versions represent the dominant systems [20]. A key element of every

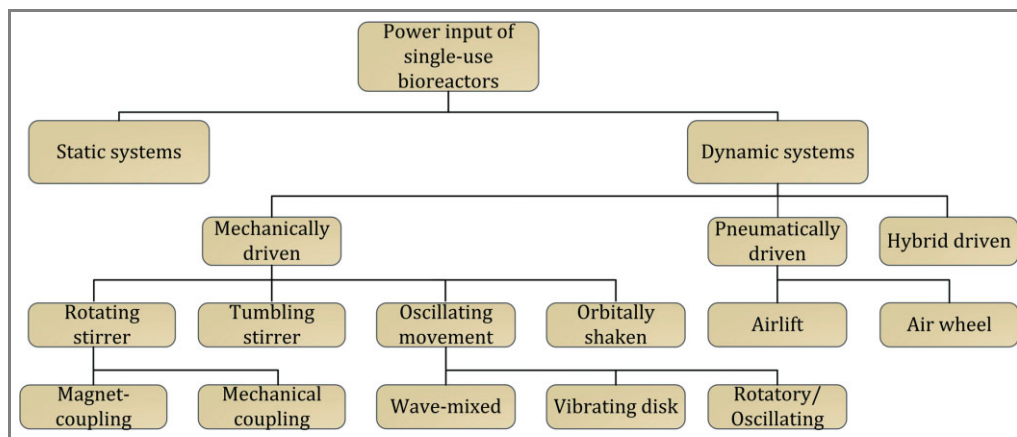


Figure 2. Categorization of single-use bioreactors based on the power input [19].

Table 1. Overview of commercially available single-use bioreactors.

Vendor	Bioreactor brand	Maximal cultivation volume	Working principle	Reactor configuration	Instrumentation	Reference
Sartorius Stedim / Merck Millipore / New Brunswick (an Eppendorf Company)	UniVessel SU / Mobius CellReady / CelliGEN BLU single-use bioreactor	2 L / 2.4 L / 50 L	Mechanically driven, rotating stirrer, mechanical coupling	Rigid vessel, cylindrical shape, static sparger	Optical and conventional pH and DO sensors	[22–24]
DASGIP (an Eppendorf Company)	DASbox single- use bioreactor	250 mL	Mechanically driven, rotating stirrer, magnetic coupling	Rigid vessel, cylindrical shape, static aeration	Conventional pH and DO sensors available	[25]
Merck Millipore / Sartorius Stedim / Xcellerex (now part of GE Healthcare)	Mobius CellReady / BIOSTAT CultiBag STR / XDR single-use bioreactor	200 L / 1000 L (2000 L is announced) / 2000 L	Mechanically driven, rotating stirrer, magnetic coupling	Flexible 3D-bag, cylindrical shape, static sparger	Optical and conventional pH and DO sensors	[26–28]
Thermo Fisher Scientific	Single Use Bioreactor, S.U.B.	2000 L	Mechanically driven, rotating stirrer, mechanical coupling	Flexible 3D-bag, cylindrical shape, static sparger	Optical and conventional pH and DO sensors	[29]
ATMI Life Sciences	Nucleo bioreactor	1000 L	Mechanically driven, tumbling stirrer	Flexible 3D-bag, cubical shape, dynamic sparger	Conventional pH and DO as well as optical density available	[30]
Finesse Solution / Sartorius Stedim / GE Healthcare	SmartBag / BIOSTAT CultiBag RM / WAVE bioreactor	25 L / 300 L / 500 L	Mechanically driven, oscillating movement (1D), wave-mixed	Flexible 2D-bag, surface aeration	Optical pH and DO sensors available	[31–33]
Applikon Biotechnology	AppliFlex	25 L	Mechanically driven, oscillating movement (1D), wave-mixed	Flexible 3D-bag, surface aeration	Conventional and single-use pH and DO available	[34]
CELLution Biotech and Lonza / Nestlé	CELL-tainer bag bioreactor / Wave and Untertow bioreactor (WUB)	150 L /250	Mechanically driven, oscillating movement (2D), wave-mixed	Flexible 2D-bag, surface aeration	Conventional pH and DO sensors available	[35, 36]
PALL Life Sciences and New Brunswick (an Eppendorf Company)	XRS 20 bioreactor system	20 L	Mechanically driven, oscillating movement (3D), wave-mixed	Flexible 2D-bag, surface aeration		[37]
Bayer Technology Service	BayShake bioreactor	1000 L	Mechanically driven, rotary oscillated movement	Flexible 3D-bag, cubical shape, static sparger or surface aeration	Optical pH and DO sensors available	[12, 21, 38]
Meissner Filtration Inc.	Vibromix (now Saltus) bioreactor	2000 L*	Mechanically driven, oscillating movement, vibrating disk	Flexible 3D-bag, cylindrical shape, dynamic sparger	Optical pH and DO sensors available	[12]

Continued Table 1.

Vendor	Bioreactor brand	Maximal cultivation volume	Working principle	Reactor configuration	Instrumentation	Reference
Sartorius Stedim	SENSOLUX Erlenmeyer flask	1 L*	Mechanically driven, orbitally shaken	Rigid shake flask, surface aeration	Optical pH and DO sensors available	[39]
TPP	TubeSpin bioreactor	400 mL	Mechanically driven, orbitally shaken	Rigid shape, surface aeration	Non-instrumented	[40]
m2pLabs / PALL Life Sciences	Biolector / Micro-24 MicroReactor	2 mL* / 7 mL	Mechanical driven, orbitally shaken	Multiwell plates, surface aeration	Optical (online) pH and DO sensors available	[41, 42]
Adolf Kühner AG and Sartorius Stedim / AmProtein	SB200-X OrbShake / CURRENT bioreactor	200 L / 300 L	Mechanical driven, orbitally shaken	Flexible 3D-bag, surface aeration	Optical pH and DO sensors available	[43, 44]
PBS Biotech	PBS bioreactor	2000 L	Pneumatically driven, air wheel	Flexible 3D-bag, cubical shape, static sparger or surface aeration	Optical pH and DO sensors available	[45]
Cellexus / Nestlé	CellMaker Regular / Slug Bubble Bioreactor (SBB)	50 L / 100 L	Pneumatically driven, airlift	Flexible 3D-bag, asymmetric shape, static sparger	Optical pH and DO sensors available	[12, 36, 46]

*total volume

single-use bioreactor (independent on the type) is the cultivation container, which is either a cylindrical rigid vessel or a flexible bag (pillow like 2D, cylindrical or cubical 3D; [21]). It is important to note that the flexible bags must be fixed and shaped by a support container or tray.

The obvious dominance of stirred single-use bioreactors is explainable by the existing bioengineering knowledge about conventional stirred systems.

3 Parametrical Methods for Single-Use Bioreactor Characterization

3.1 Flow Regime

A bioreactor's fluid flow pattern (which is, in addition to the velocity and superficial gas velocity, typically determined by parametrical methods) can be described by the Reynolds number (Re), which was introduced to describe the fluid flow in a pipe (Eq. (1)). In general, three main flow conditions can be distinguished: laminar, transient and turbulent range. A fully turbulent fluid flow pattern is achieved above a critical Reynolds number with local and time dependent fluctuations of the fluid velocity and the development of 3D swirls or eddies [8].

$$Re = \frac{\rho_L v d}{\eta_L} \quad (1)$$

For stirred bioreactors, the flow regime and the turbulence phenomena have been well investigated [47]. Therefore, the two variables for characteristic velocity and length are replaced in the basic Re equation by rotational frequency and agitator diameter (Eq. (2)). A critical Reynolds number of $Re > 10\,000$ is achieved [48, 49] when the length of the turbulent swirls is similar to both d_a and the height of the stirrer blade [6, 47]. For single-use stirred bioreactors the calculation of the Re number, the critical Re number and the determination of the length of the turbulent swirls is same as for reusable stirred bioreactors (Eq. (2)). Consequently, Eq. (2) also allows for determination of the critical Re number and, thus, the flow regime for single-use stirred bioreactors can be predicted.

$$Re = \frac{\rho_L N d_a^2}{\eta_L} \quad (2)$$

However, in the case of orbitally shaken multiwell plates, tubes, flasks [50, 51], bottles [52] or bags the fluid motion is imparted. Based on the variation of the vessel geometry, the agitator diameter is adapted to the diameter of the cultivation system. However, the diameter of the cultivation system is not constant and differs along the bioreactor height. In case of shake flasks, Büchs used the maximum inner diameter of the flask for the determination of Re number by using Eq. (2) [50, 51]. For example, turbulent conditions in

shake flasks are achieved at $Re > 60\,000$ [53]. However, for orbitally shaken systems the out-of-phase and in-phase operation phenomena have to be considered according to Büchs [50]. The bulk of the fluid circulates as the system shakes, which is referred to as being under in-phase conditions. In this case Eq. (2) can be used. If the bulk of the fluid remains only at the bottom of the cultivation system [54] and minimum fluid movement and circulation take place, this is referred to as an out of phase phenomenon [50, 55]. For rotatory oscillating bioreactors Re is simply modified and both N_s and d_a are applied, allowing for the acceleration and deceleration of the vessel motion at a defined angle [38].

The first approach, for calculating the Re number for wave-mixed bioreactors was presented in [56]. A modified Re number was introduced (Eq. (3)), with a correlation factor C ranging between 0.00747 and 0.312 (details are given in [56]). In [57] turbulent conditions above a critical Re_{mod} of 1000 were determined in the BioWave (now BIOSTAT CultiBag RM).

$$Re_{mod} = \frac{V k C \rho_L}{\eta_L (2h + B)} \quad (3)$$

3.2 Velocity Calculation

The velocity calculation refers to the maximal fluid velocity (u_{max}) generated by the agitation system. For single-use bioreactors, this parameter is influenced by the diameter of the agitator, cultivation vessel and the support container or tray dimensions. For the majority of animal cells, u_{max} should not exceed a critical value of between 1.0 and 2.0 $m\,s^{-1}$ [58] because of their shear sensitivity [59]. Nevertheless, maximum values of up to approx. 6 $m\,s^{-1}$ have been reported for hybridoma cell lines [9].

In single-use systems with rotating stirrers, the maximal introduced velocity is also referred to as tip speed (u_{Tip} ; Eq. (4)) which is also used for conventional stainless steel and glass bioreactors.

$$u_{max} = u_{Tip} = \pi N d_a \quad (4)$$

Because of the multitude of power input strategies for single-use bioreactors (see Fig. 2), the maximal induced fluid velocities must be calculated differently for wave-mixed, orbitally shaken, rotatory oscillated as well as vertically and horizontally oscillated single-use bioreactors. In the simplest case, N is modified and adapted for the system in question; as a result the maximal induced fluid velocity changes to $u_{max,mod}$. For example, the maximum velocity in the BaySHAKE bioreactor [60], which is based on the transient rotatory oscillated motion of the vessel, is calculated based on the angle velocity. By considering the angle, acceleration, deceleration and the cycle duration, an average rotary frequency was established and used with the vessel diameter to calculate the maximum velocity (Eq. (5)).

$$u_{max} = u_{max,BaySHAKE} = \pi \bar{N} d_v \quad (5)$$

However, it is not possible to apply Eq. (5) to the Vibromix bioreactor [61] because of the transient vertical upward- and downward movement of the inbuilt disk. Therefore, the maximum velocity of the vertically oscillated disk is described as a harmonic oscillation (Eq. (6)). In a scale-up process, the bioreactor and agitator dimensions should increase linearly and result in a constant u_{max} . Based on the maximum demand velocity, it is possible to calculate an average shear rate [62] that results in physiological cell death when values between 1 and 3 $\cdot 10^5\,s^{-1}$ are reached [59].

$$u_{max} = u_{max,Vibromix} = \gamma \omega_y \cos(\omega_y \, dt) \quad (6)$$

3.3 Superficial Gas Velocity

In an aerated process, the calculated superficial gas velocity (u_G , Eq. (7), [7]) is an important parameter for the prediction of dispersal (bubble/liquid) behavior in single-use bioreactors.

$$u_G = \frac{\dot{V}_G}{A} \quad (7)$$

In stirred and air lift bioreactors, it is possible to calculate the oxygen mass transfer coefficient [63]. This coefficient is only dependent on u_G for a specified medium, reactor design and sparger type combination [64]. In addition, u_G influences the bubble diameter for turbulent (churn) flow regimes [65]. However, in agitated driven bioreactors mass transfer is enhanced by agitation, which results in lower gas velocities (0.1–1 vvm; [66]) in sparged cell culture systems and reduces the requirement for addition of antifoam agents and shear protectors [67]. Furthermore, the rotational speed at which the flooding point is reached can be estimated as a constant u_G [10]. It is also worth mentioning that in stirred single-use bioreactors a relatively low agitation speed/specific power input from the pitch blade or/and segment blade impeller – which can be calculated from Eq. (4) or (5) – can result in reduced dispersal behavior and flooding of the agitator can occur in cell culture processes with typically seen aeration rates (0.1–0.2 vvm) [20, 68].

4 Experimental Methods for Single-Use Bioreactor Characterization

4.1 Power Consumption Measurement

Among the experimentally determined parameters for single-use bioreactors, which also include fluid flow pattern, mixing time, residence time, particle stress and volumetric mass transfer coefficient, the power input (P) is one of the most important and frequently used for scale-up procedures

[5, 52, 69]. If a turbulent fluid flow pattern is considered, P is responsible for the suspension of solids, the homogeneity of liquids and the dispersion of bubbles within the bioreactor liquid. In addition, the power input allows to predict the Newton number (Ne ; also called dimensionless power number), the specific power input (P/V), the power input coefficient c_p and the turbulence parameters, such as the Kolmogorov length scale [3, 7, 8].

In the case of mechanically driven bioreactors, the torque measurement (see Section 4.1.1) as well as the temperature method (see Section 4.1.2) can be applied for indirect determination of the power input. Here, the power input correlates with the rocking rate, the shaking rate, the rotation frequency of the stirrer or the cycle duration, depending on the bioreactor. For pneumatically driven bioreactors like the CELLMaker Regular, the power input is determined from the aeration rate (see [62]), whereas P/V in the PBS bioreactor depends on the rotation frequency. In cell culture processes, the P/V ranges between 10 to 250 W m^{-3} [64, 70], which is approximately 10 times lower than for microbial fermentation [9].

4.1.1 Torque Measurement

The core element for torque measurement is a torque sensor, which is installed on the moving axis and available for multiple torque ranges. For investigations of P/V in single-use bioreactors using Newtonian fluids, a torque sensor with an operating range from 1 up to 100 Nm is sufficient. However, the installation and position of the sensor within the moving axis is critical because of transverse forces exerted by the eccentrically mounted axis and its bearing. This is especially true in small scale single-use bioreactors, where torque measurement can be problematic due to the high friction of the seal, which can also be exacerbated by an inadequately installed torque sensor, resulting in the deviation of the calculated specific power input being

exceeded. In order to perform torque measurements [51, 71], the single-use bioreactor is normally filled with water and the rotation, shaking or oscillation is started. During a fixed time period and under consistent moving conditions, the torque is recorded and an average is calculated. The measurement is also carried out without any liquid in order to obtain a torque value without the weight of the liquid. Consequently, P/V can be calculated using Eq. (8).

For a comprehensive investigation of the power input in a bioreactor, the experiment should be repeated for different operating conditions. In order to interpret the results, P/V values obtained for different operating conditions are set as a function of n_{Tip} , as depicted in Fig. 3a for the BIOSTAT CultiBag STR 50L. The process should be repeated between three and five times for each set of operation conditions, to enhance accuracy and achieve a statistic average with 10 % deviation.

$$\frac{P}{V} = \frac{(M - M_d) 2 \pi N}{V} \quad \text{or} \quad \frac{P}{V} = \frac{(M - M_d) 2 \pi \bar{N}}{V} \quad (8)$$

The flow regime (see Section 3.1) can be predicted using the power characteristics. Here, the graph of the Newton number (Eq. (9)) as a function of Re (see Fig. 3b) displays the flow regime. A quasi-constant Ne value is achieved under turbulent flow regimes [8, 47]. Under these conditions, the Newton number allows the stirrer type to be identified. For example, Rushton turbines are characterized by Ne numbers between 3 and 10 [10, 72], whereas blade impellers operate at Ne number between 0.3 and 1 [72, 73]. In addition, it is well-known that impellers in unbaffled vessels show a continuous decrease in Ne number due to vortex formation [8].

$$Ne = \frac{P}{\rho_L N^3 d_a^5} \quad (9)$$

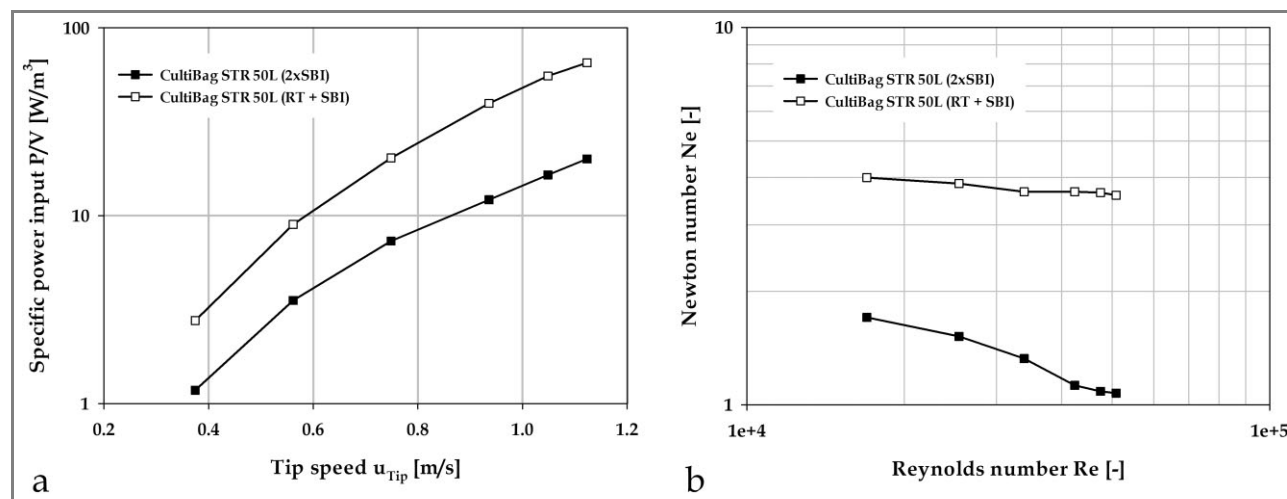


Figure 3. Specific power input as a function of tip speed (a) and the Newton number as a function of the Reynolds number (b) for the BIOSTAT CultiBag STR 50L equipped with 2 x segment-blade impeller (2xSBI) and Rushton turbine + segment-blade impeller (RT+SBI).

Due to the relatively simple installation of torque sensors in stirred, orbitally shaken and rotary oscillating single-use bioreactors, P/V data are available for these systems [6, 38, 47, 51, 52, 74–78]. In contrast, torque measurement for wave-mixed bioreactors is theoretically feasible, but there are no practical methods for the installation of a sensor. To date, no publications exist on experimental power consumption in wave-mixed single-use bioreactors.

4.1.2 Temperature Method

The temperature method is based on the energy balance of a bioreactor [52, 79], which can be recorded by temperature sensors placed in the reactor fluid and environment (Eq. (10)) [79].

$$-mC_p \frac{dT_E}{dt} = UA_H(T_E - T_0) - P \quad (10)$$

Compared to the torque method, the temperature method provides the advantage of a high-resolution measurement technique that does not measure friction from seals or mechanical devices. However, deriving the P/V is challenging [74]. This method demands the bioreactor to be insulated and requires the use of highly sensitive temperature sensors, which occasionally alter the fluid flow.

The fact that the majority of stirred, orbitally shaken and rotatory oscillating single-use bioreactors are equipped with a temperature sensor and sensor port makes the temperature method the predominant approach for determining P/V . Comparative investigations of the torque and temperature methods result in similar data for different scales of orbitally shaken single-use bioreactors that are operated with Newtonian media at temperatures between 20 and 40 °C [52, 80]. For single-use bioreactors with wave-induced motion, the temperature method provides an alternative approach for determining P/V , however, time dependent characteristics of fluid flow pattern are not taken into account.

4.2 Fluid Flow Pattern

The fluid velocity and the fluid flow pattern can be investigated by applying non-invasive methods, like laser Doppler anemometry (LDA) and particle image velocimetry (PIV), or the invasive approach of hot-wire anemometry (HWA) [81–83]. For the non-invasive methods, optical windows within the fluid are required. This is feasible for rigid plastic culture vessels, but not for flexible bag bioreactors. Hence, a replacement model with identical bag dimensions made of glass or acrylic glass is required. In this case, forming folds that occur when the bag is placed in the support con-

tainer or when the medium is put into the bag are not considered. Nevertheless, these non-invasive methods are used for the evaluation of computational fluid dynamic simulations (see Section 5). In contrast, the invasive HWA method requires no optical window. Here, the sensor has to be integrated into the fluid to measure the fluid flow pattern point by point.

In stirred single-use bioreactors, segment blade, marine or pitch blade impellers prevail, which induce an axial fluid flow [20, 70, 84]. The configuration of a segment blade impeller in combination with a Rushton turbine is possible with the BIOSTAT CultiBag STR and induces an axial as well as a radial fluid flow (see Figs. 4b and 4c). An axial fluid flow pattern was found to be generated by the eccentrically mounted pitch blade impeller in the Single-Use Bioreactor, S.U.B. [85]. In 2008, LDA investigations were performed with the Vibromix bioreactor for comparison of numerical and experimental studies [86]. Subsequently, PIV experiments were completed and an axial fluid flow pattern based on different, optimized disk geometries was observed (see Fig. 4a). For wave-mixed bioreactors the fluid velocity can be investigated using HWA [87]. The numerical results can be compared with those from experimental investigations for identical immersion depths and sensor positions [81].

4.3 Mixing Time Determination

The process of mixing is a spatial and temporal procedure and is based on the distribution of two or more components in the volume. These components differ in at least one property, such as concentration, temperature, color, density, viscosity, particle size or shape [6, 10, 88]. The component distribution depends on multiple geometric (bioreactor dimensions) and technological features (mixing quality, operation conditions and characteristics of the components, power consumption, energy dissipation) [6, 47, 87]. There-

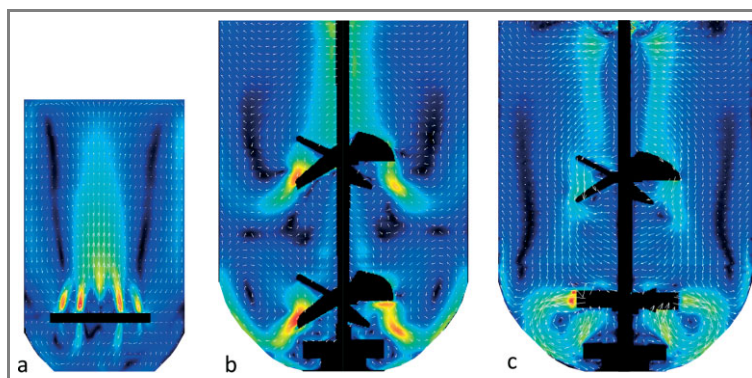


Figure 4. Velocity distribution of fluid in the Vibromix-bioreactor 15 L (a) and the BIOSTAT CultiBag STR 50L with 2 x segment-blade impeller (b) and segment-blade impeller + Rushton turbine (c) determined by PIV. Lower velocities are indicated by blue and higher velocities are indicated by red colors. The arrows show the fluid flow pattern.

fore, the mixing time (homogenization time, θ_m) is defined as the time required to achieve a certain degree of homogeneity of the elements and represents one of the most useful criteria for the characterization of bioreactors [87]. In addition, the comparison of different types of cultivation systems can be performed calculating the dimensionless mixing number (c_H) [8].

4.3.1 Decolorization Method Using Iodometry or Neutralization

The mixing time can be determined by a decolorization method that is based on a redox or neutralization reaction. This requires a dye to be put into the liquid in the bioreactor in advance. The addition of a decolorizing agent makes the color change or even disappear. The time between the addition of the decolorization solution and the entire color change or decolorization is measured and represents the mixing time for an entirely mixed fluid.

Iodometry uses a thiosulfate redox reaction with iodine and a starch indicator and is considered to be the simplest and quickest method for determining the mixing time in Newtonian fluids. The colorization is achieved through addition of an iodine-potassium iodide solution to the water-like fluid that already contains 0.2 vol% of a 1% starch solution. The two substances react and an intensive blue-black color can be observed in the fluid. By adding the decolorization agent sodium thiosulfate (with a surplus of 2.5%) the fluid is decolorized as a result of the reduction of iodine and the oxidation of thiosulfate [6]. For a neutralization reaction, sodium hydroxide and sulfuric acid with phenolphthalein as an indicator can be used. The addition of 20% sodium hydroxide solution increases the pH to 11, turning the phenolphthalein containing liquid violet. The decolorization is carried out through the addition of 50% sulfuric acid [6]. A further neutralization reaction is sodium hydroxide, hydrochloric acid and bromothymol blue as an indicator [89]. However, when tap water is used, a time delay is observed, caused by bleaching of the dye and secondary reactions [47].

The visual judgment of the decolorization process in this method leads to large deviations in results, caused by the subjectivity of the person carrying out the experiments [53]. Therefore, a parallel video recording is recommended by Carbaret [90]. Kraume and Zehner suggested repeating the mixing time determination four times [48], but in most cases it is only done in triplicate. However, the advantage of the decolorization method is that it is simple and cost-efficient and can help to localize dead zones within the bioreactor. It is worth mentioning that an optical window is required, which is provided in rigid, stirred single-use bioreactors, wave-mixed systems, the PBS Bioreactor and BIOSTAT CultiBag STR. In the case of the BIOSTAT CultiBag STR, the cylindrical part of the support container consists of two hemispherical doors, which can be opened without bursting the bag for working volumes of up to 200 L. When exceeding a working volume of 200 L the top

view can be used for decolorization methods. For single-use systems without an optical window, other methods must be used that allow mixing times to be determined by sensors and point by point.

4.3.2 Sensor Methods

To carry out a sensor method, at least one sensor that measures one physical property such as temperature or conductivity must be placed in the fluid. The mixing time is determined based on temporal changes of a certain property of an added tracer. The measurement depends on the location and type of tracer, the number, positions and response times of the sensors, as well as the sample volume and the mixing criterion [6].

The simplest sensor method to determine mixing time is the measurement of temperature gradients, thanks to the presence of temperature sensors and ports in single-use bioreactors. The addition of a solution with a different temperature to that of the reactor fluid results in a temperature shift that can be measured. Usually, the mixing time determination is performed in triplicate, however, Kraume and Zehner suggested doing it ten times for each set of operating conditions [48]. When the temperature remains constant, the fluid is homogeneously mixed in the area of the sensor. However, for highly viscous fluids the temperature method is not suitable, since the viscosity is dependent on temperature [53]. A further sensor method, and the most frequently used procedure to determine mixing time, is the conductivity method [91]. Similar in principle to the temperature method, an electrolyte solution is added to the fluid in the bioreactor and a conductivity sensor measures the change in conductivity as a function of time. For a well-mixed liquid the homogeneity can be calculated by Eq. (11) [10]. In the majority of cases, a mix of 95% homogeneity is assumed as an adequate mixing performance [3, 10].

$$H(t) = \frac{c_{T,0} - c(t)}{c_{T,0} - c_{T,\infty}} \quad (11)$$

In contrast to the decolorization method, installation of sensors within the culture vessel is required. This may alter the fluid flow pattern and make it impossible to detect dead and rest zones, since measurement is only performed for the point where the sensor is located. Increasing the number of sensors increases the accuracy with which the mixing time can be determined, but is not feasible at mL and bench top scales, and also results in deviations in measured mixing times. Furthermore, the numbers and positions of the sensors in single-use bioreactors are defined by system itself and cannot be changed by the customer. For single-use bioreactors without optical windows, the sensor methods provide a simple and practical approach for the investigation of mixing times. The sensor method was carried out for the S.U.B. [92], the XDR single-use bioreactor [27] and the BaySHAKE bioreactor [38].

4.3.3 Optical or Visual Technique

This method relies on the addition of a fluorescence dye to the bulk solution [93]. Excited by a laser, the fluorescence indicator emits light of a particular color (usually yellow-green). Fluorescence is detected by sensors or recorded by cameras [94–97]. With this technique, the mixing time is regarded as the time, when the bulk solution shows uniform color after the addition of an indicator. This application improves the accuracy of the mixing time measurement compared to that produced by the colorimetric method [53]. However, the laser induced fluorescence (LIF) requires expensive, state of the art technical equipment of PIV and an optical window within the fluid (see Section 4.2), meaning this technique is not suitable for the experimental determination of mixing time in single-use bioreactors without optical windows.

4.4 Residence Time Distribution

Residence time distribution provides a further approach for characterizing single-use bioreactors. This technique is based on the stimulus-response function when adding a tracer substance (see Section 4.3.2, [7, 8]) and is important for continuous cultivation processes [98, 99]. For a continuous process, the residence time is measured at the outlet of the bioreactor and the mean residence time is calculated by Eq. (12).

$$\bar{t} = \frac{V}{\dot{V}} \quad (12)$$

The comparison of residence time and dilution rate D indicates the quality of the fluid flow through the presence of short-circuits ($D < \bar{t}$) or dead zones ($D > \bar{t}$) [7]. Investigations using a 2-L and 20-L BIOSTAT CultiBag RM and an ideally mixed stirred bioreactor showed identical behavior. The residence time distribution shows a congruent trend [57, 100].

4.5 Particle Stress

The particle stress method is used to determine hydrodynamic exposures in a defined substance system. For this purpose, so-called floc or emulsion systems are used, which are measurably affected by normal and shear stress [49]. Such hydrodynamic forces are responsible for damaging animal cells in expansion or protein production processes. Therefore, understanding these forces is vitally important to keep local shear stress in a range that cells can tolerate.

The standard floc system uses distilled water, Witterschlicker Blauton HFF and the flocculant Praestol 650 BC [38, 49, 71, 101, 102]. In the case of an emulsion system, distilled water, a synthetic oil (e.g. Mobil EAL Arctic 22; [49]) and a non-ionic surfactant (e.g. Triton X-100) are the main components [49]. For the experimental determination of particle stress, a focused beam reflectance measurement

(FBRM) sensor is needed, which is integrated in the vessel at an angle of 45° and detects the mean chord length distribution as a function of time [71].

The reference floc diameter (d_{VF}) is used for the comparison of different bioreactors. d_{VF} is reached at a decay rate of $d_{VF}/dt = -0.33 \mu\text{m min}^{-1}$ and is directly proportional to the mean chord length [103, 104]. The diameter of droplets in emulsions is constant after a certain time depending on experimental conditions. The graphs of floc and droplet diameter as a function of P/V are used to analyze different power input conditions and bioreactor scales. In spite of the expense of the measurement technique, this method is well suited for the determination of hydrodynamic effects in the substance system, and for single-use bioreactors running below a P/V of 10 W m^{-3} [49]. Investigations of particle stress in single-use systems have only been carried out in the BayShake bioreactor under varying operating and aeration conditions, using the described floc system. For a P/V of 15 up to 120 W m^{-3} a decrease in d_{VF} of 200 to $25 \mu\text{m}$ was measured [38], indicating no hydrodynamic effects on CHO cells with diameters ranging between 10 and $20 \mu\text{m}$.

Further investigations are provided by Kalmbach et al. [81], who used an especially designed hot-film sensor and a special flush-mounted film sensor to determine shear stress and wall shear stress in a 2-L and 20-L wave-mixed bioreactor. Here, lower shear stress was found in the cultivation bag compared to conventional stirred bioreactors [81].

4.6 Volumetric Mass Transfer Coefficient

In aerobic cultivation processes, the supply of oxygen is provided by spargers, membranes, open pipes or aeration of the fluid surface [105, 106]. The transfer of the oxygen from the gas bubble into the liquid phase is described by the two-film or penetration theory and defines the oxygen transfer rate (OTR [107]). The OTR depends on the aeration device, the mixing efficiency, the power input, the aeration rate and the fluid properties [63]. In the presence of cells, the dissolved oxygen (DO) is consumed by the biomass, a process that is described as the oxygen uptake rate (OUR) from the liquid into the biomass and allows the specific physiological nature of the culture to be described [108]. The accumulation rate of oxygen in the culture broth can be written as the difference between OTR and OUR and is provided in Eq. (13).

$$\frac{dc}{dt} = \text{OTR} - \text{OUR} = \left[k_L a \left(c_{L,O_2}^* - c_{L,O_2} \right) \right] - [q_{O_2} c_X] \quad (13)$$

However, the mass transfer coefficient and specific interfacial area are not easy to determine and thus both parameters are combined to give the volumetric mass transfer coefficient $k_L a$ [108], which is defined as the reciprocal time of oxygen transfer from gas to liquid phase, assuming ideally mixed fluids. In addition to the multitude of empiri-

cal approaches for the calculation of $k_L a$ in bioreactors [107, 109–112], a large number of different experimental methods has been published for the direct and embedded determination of $k_L a$. In order to achieve statistical accuracy, all investigations should be executed in triplicate with a deviation of up to 10%. According to Section 4.3.2, the sensor measurement depends on the location, number and position, which is influenced by the $k_L a$ value and should be described for the bioreactor and method. The response time of the sensor must also be determined and considered as part of the $k_L a$ calculation. In cell culture processes, knowledge of $k_L a$ is tremendously important for an adequate aeration strategy because of changes in cell metabolism or cell damage caused by sustained oxygen limitation, and the $k_L a$ value should be in range between 1 and 15 h⁻¹ [64, 70, 113]. Additionally, high gas transfer rates promote dense foam formation and shear sensitive cells can be injured by the bursting of gas bubbles [67]. In the following sections examples of dynamic methods, which are defined by changes in the dissolved oxygen concentration during the measurement time [10, 47] are described. Furthermore, a static method is also described (static sulfite method), where the dissolved oxygen concentration is zero during the measurement time and over the balance area of the bioreactor [47]. This method measures the dissolved oxygen indirectly through the concentration of a further reagent.

4.6.1 Gassing-Out Method

In the literature the gassing-out method, which is also referred to as the dynamic-gassing method [63], static gassing-out method [114] or dynamic method [107] is the predominant method for the determination of $k_L a$. Because of the inconsistency in the terms used to describe this method, the fact that nitrogen is used to eliminate the air, and the accumulation of dissolved oxygen, the subsequently described method will be called the gassing-out method.

The gassing-out method is based on the measurement of the DO in the absence of biomass and thus does not take the OUR (OUR = 0 mol O₂ m⁻³ s⁻¹; Eq. (13)) into account. Phosphate buffered saline (PBS), 0.5 M sodium sulfate (Na₂SO₄), distilled water or process media are most often used to carry out this procedure. The method eliminates the oxygen in the fluid by introducing nitrogen until the DO concentration is equal to zero. Oxygen or air is subsequently added until the liquid is saturated with DO [63, 107]. The elimination and saturation of oxygen is calculated by Eqs. (14) and (15). The analysis of $k_L a$ is performed by means of a graphical representation with the logarithmical concentration of DO as a function of time. Here, the slope between 10 and 90% of DO represents the $k_L a$ value.

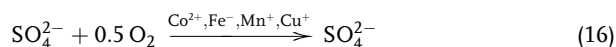
$$\ln\left(\frac{C_L}{C_{L,O2}}\right) = k_L a t \quad (14)$$

$$\ln\left(1 - \frac{C_{L,O2}}{C_{L,O2}^*}\right) = -k_L a t \quad (15)$$

The gassing-out method is a simple and cost-effective method that is frequently mentioned in the literature as the method of choice for the determination of $k_L a$ in standard stirred and single-use bioreactors such as the UniVessel SU [115], the Mobius CellReady [116], the CultiBag STR [105] and the S.U.B. [92]. Investigations with the gassing-out method have already been done for rotatory oscillated [38], orbitally shaken [117] and wave-mixed single-use bioreactors [118, 119]. Furthermore, a comparison of different single-use and standard stirred bioreactors is feasible by uniformly determining $k_L a$. However, the application of distilled water results in divergent $k_L a$ values. Culture media are rarely used because of their high costs. For comparative studies a salty medium such as 1% PBS or 0.5 M Na₂SO₄ is preferred.

4.6.2 Sulfite Methods

The dynamic sulfite method is based on the chemical reaction of sulfite (SO₃²⁻) to sulfate (SO₄²⁻) under the presence of DO (Eq. (16)). The reaction is catalyzed by copper, ferric, cobalt or manganese ions.



Determining $k_L a$ using the dynamic sulfite method [120] is performed in distilled water with one of the aforementioned catalysts (mostly Co²⁺). The catalyst concentration depends on the Hatta number (Ha), which should be in the non-accelerated range (Ha < 0.3) [120]. After the addition of sodium sulfite to the fluid (with a surplus of 20%) the DO decreases immediately. Oxygen consumption stops after the sulfite has been reduced completely and the minimal DO has been reached. The subsequent aeration leads to an increase in DO up to the maximum DO. A saturation curve is generated and the $k_L a$ can be calculated using Eq. (15) (see Section 4.6.1). In the static sulfite method [121] the same principle of oxygen reduction is used, as described in Section 4.6.1. Here, the concentration of sulfite is used to determine $k_L a$ and no saturation curve is recorded. After the DO is consumed, the air is turned on allowing the oxidation reaction to continue for some minutes. After stopping the air flow and agitating, a sample of known volumes is taken at regular time intervals. Subsequently, the sample is mixed with an excess of standard iodine reagent and finally titrated with standard sodium thiosulfate solution to a starch indicator end point [107].

The dynamic and static sulfite methods provide good alternatives to the gassing-out method (see Section 4.6.1), with simple procedures that requires minimal expenditure of time [117]. In addition, a maximum deviation of 10% was found for orbitally shaken bioreactors (TubeSpin, Shake

flask and orbitally shaken bags) when comparing both methods [117]. Furthermore, the static sulfite method even allows the $k_L a$ to be determined at mL-scale. When using an indicator ($\text{pH} \neq \text{const.}$), the time it takes for stoichiometric sulfide to oxidize can be measured, thanks to the pH variation or the color change of the indicator, which can be recorded by a camera [120, 121]. Finally, the absence of a dispersed gas phase is an ideal starting point for the application in single-use bioreactors with surface aeration [122], because of the small bubble effect inhibition, which should result in a correct determination of the $k_L a$ value [120]. However, the cobalt solution that is used in this procedure is hazardous waste and must be disposed of separately.

4.6.3 Respiratory Gassing-Out Method

In 1967, Bandyopadhyay et al. developed the respiratory gassing-out method, where the $k_L a$ is determined in presence of respiratory biomass [123]. In the literature, the method is also called the dynamic method or the direct dynamic method [107, 124–128]. The procedure is similar to that of the gassing-out method (see Section 4.6.1), but also considers the OUR (desorption). For the purposes of this review, the method will be referred to as the respiratory gassing-out method.

In respiratory active cultures, desorption is induced when aeration is stopped. The DO then decreases at the same rate as the biomass respires the DO (Eq. (17)). For a known cell concentration, the OUR can be determined by the graphical slope of DO as a function of time. When a critical oxygen concentration is reached, the aeration is turned on again, the DO increases until a steady state concentration is achieved. To determine the $k_L a$, one has to consider the cell concentration and calculate the value using the slope of DO concentration as a function of time (Eq. (18)). For more information of the respiratory method, a detailed mathematical derivation is provided by Garcia-Ochoa [107].

$$\frac{dC_L}{dt} = \text{OUR} = -q_{O_2} c_x \quad (17)$$

$$q_{O_2} c_x \Delta t + \Delta c_{L,O_2} = k_L a (c_{L,O_2}^* - c_{L,O_2}) \Delta t \quad (18)$$

The respiratory gassing-out method presents a reliable and realistic method to determine the $k_L a$ value, because it is determined within the process and under cultivation conditions. However, to compare different bioreactors, process and cultivation conditions must be identical in every process run, something that is currently not feasible. In addition, $k_L a$ influencing parameters like power input, aeration, temperature, etc., have to be in the same range as the real cultivation conditions. Furthermore, cell culture medium is often supplemented with non-ionic surfactants

such as Pluronic, which is a cell protection agent and is associated with its ability to suppress cell attachment to bubble surface and the surface tension [59]. The reduction of surface tension results in decreased $k_L a$ values [129–131].

Besides the OTR influencing parameters (physical and chemical) the OUR (Eq. (13)) or the specific oxygen uptake rate (q_{O_2}) is also influenced by the biomass type, its concentration, the proliferation state of the biomass and the concentration of essential metabolites such as glucose or glutamine [67]. As a result of this fact, biomass growth is considered to be an altering process. The high costs of cell culture media and the determination of the $k_L a$ in an aseptic process, which also increases the risk of contamination, can be considered as additional disadvantages of this method. Due to the various physical, chemical and biological factors that influence the process, the respiratory gassing-out method is not suggested for characterizing single-use bioreactors. But the method can be used for characterization of cell culture-based processes.

5 Computational Fluid Dynamics as Numerical Approach

5.1 Fluid Flow Pattern and Velocity Distribution

CFD is a computer-aided method to describe local and time resolved fluid flow patterns and includes three main steps: (1) the pre-processing, (2) processing and (3) post-processing. Pre-processing includes the construction of the required bioreactor geometry, using computer-aided design and mesh generation. Furthermore, boundary and initial conditions must be defined in order to solve the differential equations [84]. The differential equations are based on the equations describing the conservation of mass, momentum and energy (processing) [132–137]. Because CFD is a numerical approach, rigorous experimental validation is essential (post-processing). Analytical techniques, such as PIV or LDA have recently been developed (see Section 4.2), enabling the accurate measurement of the local fluid velocity. But in all cases, the results obtained through simulation must be validated and if possible verified by further experimental investigation. Based on CFD, the calculation of the velocity distribution and the fluid flow pattern can be predicted (see Fig. 5). In a following step the power input can be calculated using the force or moment. This numerical technique is similar to the torque method (see Section 4.1.1). However, by using CFD, both the number of single-use bioreactor prototypes and the associated qualification experiments can be reduced to a minimum. This will finally result in cost and time savings in development, characterization and manufacturing. For more detailed information about CFD investigations focusing on fluid flow patterns and velocity distribution in single-use bioreactors refer to [20, 84–86, 138].

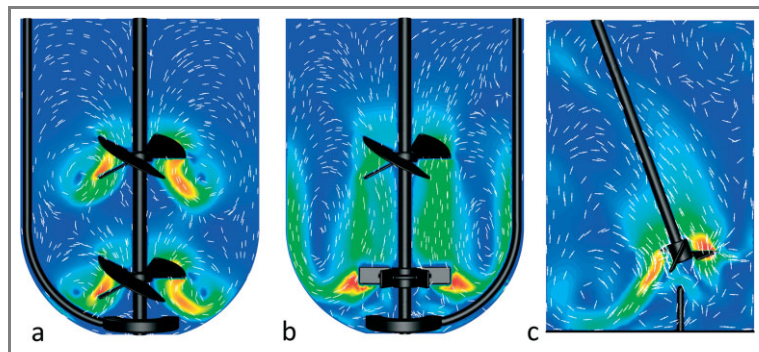


Figure 5. Numerical predicted velocity distribution of the fluid in the BIOSTAT CultiBag STR 50L with segment-blade impeller + Rushton turbine (a) and 2 x segment-blade impeller (b) as well as the S.U.B. 50L (c). Lower velocities are indicated by blue and higher velocities are indicated by red. The arrows show the fluid flow pattern.

5.2 Mixing Time Determination

The simulation of a mixing process is based on a steady flow field. This is followed by a transient simulation using a disabled velocity equation (the velocity distribution is frozen), allowing to determine the concentration distribution of an inert tracer. The transport equation of the convective and diffuse tracer distribution is given in [139].

The numerical determination of mixing time is similar to the conductivity method, where a concentrated fluid is added to the liquid in the reactor and the changes in concentration are measured by at least a conductivity sensor (see Section 4.3.2). In contrast, when using CFD, sensor measurements can be determined for any of the control volumes resulting in a high-resolution mixing time prediction. For numerical mixing time, a tracer with identical fluid properties is added with a maximum volume of 1 % of the total volume. Therefore, the effect of volume change is negligible. The tracer concentration is recorded in significant areas, for example, where the fluid velocities are very low or decrease to the minimum. The generated plots of the significant areas look like the experimental graphs, produced by the conductivity or calorimetric methods. In addition, recording of the volume integral of the minimum and maximum tracer concentrations is required. Thus, inhomogeneities in the fluid domain are excluded. The mixing time is given when the concentrations of the points and the volume integrals are constant. At this particular time, there is a homogeneous tracer concentration in the vessel. Detailed investigations and the numerical approach are provided by [85, 140].

5.3 Volumetric Mass Transfer Coefficient

The multiphase simulation approach increases the computational effort because of the additional conservation equations for mass, energy and momentum for each phase. Furthermore, models describing the interaction between the two phases must also be considered [132–135, 141].

Based on a constant bubble size, the numerical determination of $k_L a$ is possible in steady or transient solver mode [20]. When using the population balance model, which includes a calculation for bubble break up and coalescence, the bubble size distribution can only be calculated in a transient mode and using the $k-\epsilon$ model. Therefore, a velocity inlet and a pressure outlet have to be set for boundary conditions. During the simulation, the concentration of the gas phase must be recorded. If the concentration of the gas phase is constant, the maximum dissolved oxygen concentration in the fluid has been reached and the volumetric mass transfer coefficient can be determined in a similar fashion to the gassing-out method (see Section 4.6.1). Detailed investigations and the numerical approaches are provided in [20, 138].

5.4 Energy Dissipation

A novel approach for the comparison of different bioreactors is described by Werner [142, 143]. The relative volume as a function of the spatial distribution of the specific turbulent energy dissipation rate is used. For a homogeneous distribution of dissipation rate, the slope of the curve becomes steeper. It was found, that BIOSTAT CultiBag RM (2 and 200 L scale) exhibits a more homogenous energy dissipation when compared to 2-L reusable stirred bioreactors equipped with a Rushton turbine and a paddle impeller. Detailed investigations are provided by Löffelholz [84].

6 Conclusions

Cultivation in single-use bioreactors requires fundamental knowledge of the applied reactor system, the cells that are to be grown and how to control the bioprocess. Determination of the flow regime and fluid velocity, investigation into the power consumption (torque or temperature method), mixing time (decolorization or sensor method) and $k_L a$ (gassing-out or sulfite method) represent the major engineering methods used to characterize single-use bioreactors. Tab. 2 summarizes the methods recommended for determining the main engineering parameters for single-use bioreactors. The methods consider power consumption, shape and configuration of the bioreactor cultivation vessel, its instrumentation and scale.

In the case of rigid cultivation systems, it is recommended to determine the flow regime, fluid velocity and power consumption using torque measurement, since agitation is effected through a moving axis. In orbitally shaken systems with large volumes, the temperature method can be applied instead to determine the power consumption. In contrast, the investigation of mixing time should only be performed using a fluid color change, since optical windows are

Table 2. Methods for determining the main engineering parameters of single-use bioreactors (recommended = x).

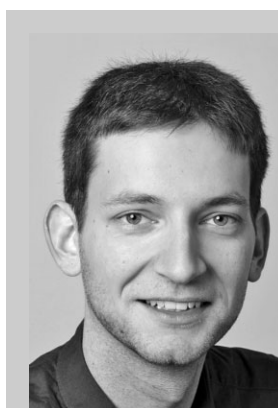
			Rigid vessels		Flexible bags		
			mL-scale	L-scale	wave-mixed / orbitally shaken	tumbling stirrer	rotating stirrer / oscillating movement (rotary and vibrating)
Parametrical methods	Flow regime		x	x	x	x	x
	Fluid velocity		x	x	x	x	x
	Superficial gas velocity		x	x		x	x
Experimental methods	Power consumption	Torque method	x	x	x	x	x
		Temperature method		x ³			
	Mixing time	Decolorization	x	x	x	x	x ¹
		Probe methods					x ²
	$k_L a$	Gassing-out method	x ⁴	x	x	x	x
		Sulfite method	x ³	x ³	x ³		

1) Only for single-use bioreactors with optical window; 2) for single-use bioreactors without optical window; 3) optional for orbitally shaken and wave mixed; 4) if optical sensors available.

available and these methods provide the possibility to localize dead and rest zones. Depending on the bioreactor volume, the sulfite (mL-scale) or gassing-out method (larger scale) should be preferred for determining $k_L a$. However, in single-use bag bioreactors the torque and gassing-out methods provide the most meaningful approaches for determining the power consumption and $k_L a$. In the case of wave-mixed bioreactors, tumbling stirrers, air wheels and the BIOSTAT Culti-Bag STR, the decolorization method represents the most suitable method for investigating mixing time, since the fluid is optically accessible. For other flexible single-use systems like the S.U.B., sensor methods are recommended because this approach only allows punctual measurements. In addition, the residence time must be determined when using continuously operating single-use systems.

An advanced engineering characterization of single-use bioreactors should include the experimental determination of particle stress or the implementation of CFD in order to obtain information about how flow can influence a system. Moreover, it allows time and spatial resolved fluid flow patterns to be observed. However, results obtained by CFD must be evaluated and verified through experimental determination of fluid flow behavior.

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Remark

This paper was initiated by the upstream processing group of the temporary working committee of DECHEMA "Single-Use Technology in Biopharmaceutical Manufacturing". The group intends to provide a catalogue of methods, containing detailed procedure descriptions, for the characterization of

the main important single-use bioreactors recommended in Tab. 2.

Symbols used

a	[m ⁻¹]	specific interfacial area
A	[m ²]	area
A_H	[m ²]	heat transfer area
B	[m]	bag width
c	[kg m ⁻³]	concentration
c_H	[-]	dimensionless mixing number
c_P	[-]	dimensionless power input coefficient
c_L	[kg m ⁻³]	dissolved oxygen concentration in the liquid
c_{L,O_2}^*	[kg m ⁻³]	saturation concentration of the oxygen in the liquid
$c_{L,O_2}^* - c_{L,O_2}$	[kg m ⁻³]	difference of dissolved oxygen in the fluid
c_T	[kg m ⁻³]	tracer concentration
$c_{T,0}$	[kg m ⁻³]	tracer concentration at the beginning of the determination
$c_{T,\infty}$	[kg m ⁻³]	tracer concentration at the end of the determination
c_x	[cells mL ⁻¹]	cell concentration
C_p	[kg s ⁻² m ⁻¹ K ⁻¹]	heat capacity
C	[-]	correlation factor
d	[m]	characteristic length
d_a	[m]	agitator diameter
d_v	[m]	vessel diameter
d_{VF}	[m s ⁻¹]	reference floc diameter
D	[s ⁻¹]	dilution rate
h	[m]	high level of the bag
H	[-]	homogeneity
Ha	[-]	Hatta number
k_L	[m s ⁻¹]	mass transfer coefficient
$k_L a$	[s ⁻¹]	volumetric mass transfer coefficient
m	[kg]	mass
M	[kg m ² s ⁻²]	torque
M_d	[kg m ² s ⁻²]	dead weight torque
Ne	[-]	Newton number
N	[s ⁻¹]	rotation frequency
$\bar{N}$	[s ⁻¹]	average rotary frequency
OTR	[kg O ₂ m ⁻³ s ⁻¹]	oxygen transfer rate
OUR	[kg O ₂ m ⁻³ s ⁻¹]	oxygen uptake rate
P	[kg m ² s ⁻³]	power input
P/V	[kg m ⁻¹ s ⁻³]	specific power input
q_{O_2}	[kg O ₂ s ⁻¹ cells ⁻¹]	specific oxygen uptake rate
r	[s ⁻¹]	rocking rate
Re	[-]	Reynolds number
t	[s]	time
$\bar{t}$	[s]	mean residence time
T_E	[K]	fluid temperature

T_0	[K]	environment / bioreactor wall temperature
u_G	[m s ⁻¹]	superficial gas velocity
u_{max}	[m s ⁻¹]	maximal fluid velocity
$u_{max, mod}$	[m s ⁻¹]	modified maximal fluid velocity
u_{Tip}	[m s ⁻¹]	tip speed
U	[kg K ⁻¹ s ⁻³]	heat transfer coefficient
v	[m s ⁻¹]	characteristic velocity
V	[m ³]	working volume
$\dot{V}$	[m ³ s ⁻¹]	flow-rate (liquid)
$\dot{V}_G$	[m ³ s ⁻¹]	flow-rate of gas
γ	[m]	amplitude

Greek symbols

η_L	[kg m ⁻¹ s ⁻¹]	fluid viscosity
θ_m	[s]	mixing time
ρ_L	[kg m ⁻³]	fluid density
ω_y	[s ⁻¹]	angle velocity

Abbreviations

CFD	computational fluid dynamics
CHO	Chinese hamster ovary
DO	dissolved oxygen
FBRM	focused beam reflectance measurement
FDA	Food and Drug Administration
HWA	hot-wire anemometry
LDA	laser Doppler anemometry
LIF	laser induced fluorescence
NSE	Navier-Stokes equations
PBS	phosphate buffered saline
PIV	particle image velocimetry

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